

Classification of Algebraic Varieties I

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Outline of the talk

1 Introduction

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- 1 Introduction
- 2 Irreducible subsets

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- 3 Curves

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- 2 Irreducible subsets
- 3 Curves
- 4 surfaces

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- 2 Irreducible subsets
- 3 Curves
- 4 surfaces
- 5 Higher dimensions. Preview.

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Polynomial equations

Algebraic Geometry is concerned with the study of solutions of polynomial equations say n -equations in m -variables

$$\begin{cases} P_1(x_1, \dots, x_m) = 0 \\ \dots \\ P_n(x_1, \dots, x_m) = 0. \end{cases}$$

1 equation in 1 variable

- For example, 1 equation in 1 variable

$$P(x) = a_d x^d + a_{d-1} x^{d-1} + \dots + a_1 x + a_0 \quad a_d \neq 0$$

(a polynomial of degree d in x).

- If $a_i \in \mathbb{C}$, then by the Fundamental Theorem of Algebra, we may find $c_i \in \mathbb{C}$ such that $P(x)$ factors as

$$P(x) = a_d (x - c_1)(x - c_2) \cdots (x - c_d).$$

- Therefore a polynomial of degree d always has exactly d solutions or roots (when counted with multiplicity).
- If one is interested in solutions that belong to $\mathbb{R}$ (or $\mathbb{Q}$, or $\mathbb{Z}$ etc.), then the problem is much more complicated, but at least we know that there are at most d solutions.

Complex solutions

Throughout this talk I will always look for complex solutions $(z_1, \dots, z_m) \in \mathbb{C}^m$ to polynomial equations

$$P_1(x_1, \dots, x_m) = 0 \quad \dots \quad P_n(x_1, \dots, x_m) = 0$$

where $P_i \in \mathbb{C}[x_1, \dots, x_m]$.

Projective space

- It is more convenient to work on a compactification $\mathbb{C}^N \subset \mathbb{P}_{\mathbb{C}}^N$ where $\mathbb{P}_{\mathbb{C}}^N = \{\mathbb{C}^{N+1} - (0, \dots, 0)\} / \{\mathbb{C}^*\}$ is projective N -space.
- We have that $\mathbb{P}_{\mathbb{C}}^0$ is a point, $\mathbb{P}_{\mathbb{C}}^1 \cong S^2$ is the Riemann sphere and $\mathbb{P}_{\mathbb{C}}^N = \mathbb{C}^N \cup \mathbb{P}_{\mathbb{C}}^{N-1}$.
- We then replace $P(x_1, \dots, x_N)$ by the corresponding homogeneous polynomial in $\mathbb{C}[x_0, \dots, x_N]$. Eg.
 $x_1^2 + x_2^3 \Rightarrow x_0 x_1^2 + x_2^3$.
- We then have:

Bezout's Theorem

Theorem (Bezout's Theorem)

Let $P_1(x_0, \dots, x_n), \dots, P_n(x_0, \dots, x_n)$ be n homogeneous polynomials of degrees $d_1, \dots, d_n$. If the set

$$\bigcap_{i=1}^n \{P_i(x_0, \dots, x_n) = 0\} \subset \mathbb{P}_{\mathbb{C}}^n$$

is finite, then it consist of exactly $d_1 \cdots d_n$ points (counted with multiplicity).

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Irreducible subsets

- In general a **Zariski closed subset** of $\mathbb{P}_{\mathbb{C}}^m$ is a set of the form

$$X = \bigcap_{i=1}^n \{P_i(x_1, \dots, x_m) = 0\} \subset \mathbb{P}_{\mathbb{C}}^m.$$

(Where $P_1, \dots, P_n \in \mathbb{C}[x_0, \dots, x_m]$ are homogeneous polynomials.)

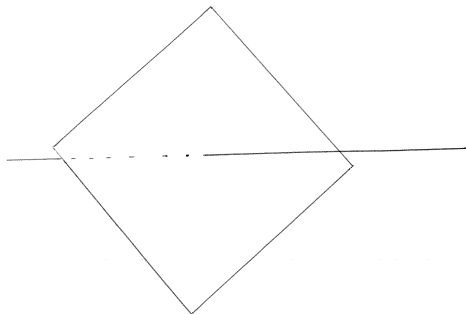
- A Zariski closed subset of $\mathbb{P}_{\mathbb{C}}^m$ is **irreducible** if it can not be written as the union of 2 proper Zariski closed subsets.
- Any Zariski closed subset is a finite union of irreducible Zariski closed subsets.
- If $P(x, y, z) \in \mathbb{C}[x, y, z]$ is homogeneous of degree $r > 0$, then

$$X = \{P(x, y, z) = 0\}$$

is irreducible if and only if $P(x, y, z)$ does not factor.

Reducible set

REDUCIBLE ZARISKI CLOSED SUBSET OF $\mathbb{P}^3_{\mathbb{C}}$

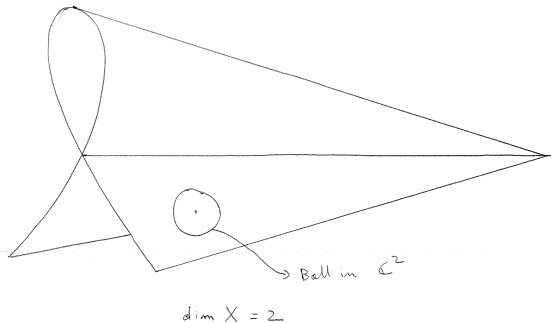


Irreducible subsets II

- If X is an infinite irreducible Zariski closed subset, then there is an integer $d > 0$ such that for most points $x \in X$, one can find an open neighborhood $x \in U_x \subset X$ where U_x is analytically isomorphic to a ball in $\mathbb{C}^d$.
- We then say that x is a **smooth** point and that X has **dimension** d .
- By a deep result of Hironaka, there exists a **desingularization** of X , that is a morphism $f : Y \rightarrow X$, such that Y is everywhere smooth and f is the identity over any smooth point of X .

Irreducible set of dimension 1

SINGULAR SET IN $\mathbb{P}^3_{\mathbb{C}}$



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- 1 Introduction
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- 3 **Curves**
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Curves

- A **curve** is an irreducible Zariski closed subset of dimension 1.
- All smooth curves are orientable compact manifolds of real dimension 2 and so they are Riemann Surfaces.
- Topologically, a curve X is determined by its genus.
- The simplest curves are **rational curves** i.e. curves of genus 0.
- There is only one such curve $X = \mathbb{P}_{\mathbb{C}}^1$.
- When $g = 1$ we have a 1-parameter family of **elliptic curves** $X_t = \{y^2 - x(x-1)(x-t) = 0\}$.

Curves of general type

- If X is a curve with genus $g \geq 2$ then we say that X is a curve of **general type**.
- There is a $3g - 3$ -parameter family of curves of general type.
- For any curve X there are (infinitely) many different descriptions

$$\bigcap_{i=1}^n \{P_i(x_1, \dots, x_m) = 0\} \subset \mathbb{P}_{\mathbb{C}}^m.$$

- We would like to find a natural (canonical) description for $X \subset \mathbb{P}_{\mathbb{C}}^m$.

Line bundles

- Since X is compact, the only global holomorphic functions are constant. Therefore, we consider meromorphic functions with prescribed poles or equivalently sections of line bundles.
- If L is a line bundle on X and $s_0, \dots, s_m$ is a basis of $H^0(X, L)$, then there is a map $\phi_L : X \dashrightarrow \mathbb{P}_{\mathbb{C}}^m$ given by

$$x \rightarrow [s_0(x) : \dots : s_m(x)]$$

(undefined at the common zeroes of the s_i).

- To define an embedding of X in $\mathbb{P}_{\mathbb{C}}^m$, we must find a line bundle L whose sections distinguish different points and tangent directions.

The canonical line bundle

- There is essentially only one natural choice: the **canonical line bundle** ω_X and its tensor powers $\omega_X^{\otimes n}$.
- The sections of $H^0(X, \omega_X^{\otimes n})$ can locally be written as $f(x)dx^{\otimes n}$.
- We have $\omega_{\mathbb{P}^1} = \mathcal{O}_{\mathbb{P}^1}(-2)$ so that $H^0(\omega_{\mathbb{P}^1}^{\otimes n}) = 0$ for $n > 0$.
- When $g(X) = 1$, we have $\omega_X = \mathcal{O}_X$ so that $H^0(\omega_X^{\otimes n}) = \mathbb{C}$ for $n \geq 0$.

Theorem

Let X be a curve of genus $g \geq 2$, then for any $k \geq 3$ the sections of $H^0(X, \omega_X^{\otimes k})$ define an embedding

$$X \hookrightarrow \mathbb{P}_{\mathbb{C}}^{(2k-1)(g-1)-1}.$$

The canonical ring

- There is a more abstract/natural point of view:
- Define the **canonical ring**

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k}).$$

- If $g = 0$, $R(\omega_X) \cong \mathbb{C}$ and if $g = 1$, $R(\omega_X) \cong \mathbb{C}[t]$.
- If $g \geq 2$, then $\omega_X^{\otimes 3} = \mathcal{O}_{\mathbb{P}^N}(1)|_X$. It follows that $R(X)$ is finitely generated $\mathbb{C}$ -algebra which captures the geometry of X . In fact

$$X = \text{Proj } R(\omega_X).$$

- In coordinates this may be described as follows. If $r_0, \dots, r_m$ are generators of $R(\omega_X)$ and $K = \ker(\mathbb{C}[x_0, \dots, x_m] \rightarrow R(\omega_X))$, then $X \subset \mathbb{P}_{\mathbb{C}}^m$ is defined by the equations in K .

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The canonical ring

- Let X be a smooth variety of dimension d . The sections $s \in H^0(X, \omega_X^{\otimes k})$ may be locally written as $f(x_1, \dots, x_d)(dx_1 \wedge \dots \wedge dx_d)^{\otimes k}$.
- The **Canonical Ring** of X is given by

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k}).$$

- The **Kodaira dimension** of X is $\kappa(X) = \text{tr.deg.}_{\mathbb{C}} R(\omega_X) - 1$.
- We have $\kappa(X) = \max_{k \geq 0} \{\dim \phi_{\omega_X^{\otimes k}}(X)\}$ so that $\kappa(X) \in \{-1, 0, 1, \dots, d\}$.

Birational geometry of surfaces

- The classification of (smooth) surfaces ($\dim(X) = 2$) was already understood by the Italian school of Algebraic Geometry around the beginning of the 20th-century.
- The first problem in the classification of surfaces is that given any point on a surface $x \in X$, one can produce a morphism

$$\nu : \tilde{X} = \text{Bl}_x(X) \rightarrow X$$

known as the **blow up** of X at x .

- The morphism ν may be viewed as a surgery that replaces the point x by a rational curve $E \cong \mathbb{P}^1 = \mathbb{P}(T_x X)$.
- The exceptional curve E is called a -1 **curve** since

$$\omega_{\tilde{X}} \cdot E = \deg(\omega_{\tilde{X}}|_E) = -1.$$

Birational geometry of surfaces II

- We say that two surfaces are **birational** if they have isomorphic open subsets (Zariski topology).
- This is a new phenomenon: two curves are birational if and only if they are isomorphic.
- It turns out that two surfaces X and X' are birational if and only if there is a sequence of blow ups $X_n \rightarrow X_{n-1} \rightarrow \dots X_1 \rightarrow X$ and $X'_m \rightarrow X'_{m-1} \rightarrow \dots X'_1 \rightarrow X'$ such that $X_n = X'_m$.
- It is then natural to study surfaces up to birational equivalence.

Birational geometry of surfaces III

- By a Theorem of Castelnuovo, one may reverse the blowing up procedure i.e. given any surface X and any -1 curve $E \subset X$, there exists a morphism $\nu : X \rightarrow X_1$ such that $X = \text{Bl}_x(X_1)$ for some point $x \in X_1$.
- Note that $\rho(X_1) = \rho(X) - 1 \geq 0$ and so this procedure can be repeated at most finitely many times.
- A **minimal surface** is a surface that contains no -1 curves.
- By Castelnuovo's Theorem, given any surface X there exists a birational morphism to a minimal surface $X \rightarrow X_{\min}$ (given by finitely many point blow downs).

Birational geometry of surfaces IV

- If $\kappa(X) = -1$, then the minimal model $X \rightarrow X_{\min}$ is not unique (but it is well understood how two different minimal models are related).
- If $\kappa(X) = -1$, then X is covered by rational curves (i.e. by $\mathbb{P}^1$'s). The sections of $H^0(\omega_X^{\otimes m})$ must vanish along these curves for $m > 0$. Hence $R(\omega_X) \cong \mathbb{C}$.
- If $\kappa(X) \geq 0$, then the minimal model $X \rightarrow X_{\min}$ is unique and $\omega_{X_{\min}}$ is **nef**. This means that for any curve $C \subset X_{\min}$ we have

$$\omega_{X_{\min}} \cdot C \geq 0.$$

- In fact one can show that $\omega_{X_{\min}}$ is **semiample** i.e. there is an integer $m > 0$ such that the sections of $H^0(\omega_{X_{\min}}^{\otimes m})$ have no common zeroes.

Birational geometry of surfaces V

- Therefore, $\phi_{\omega_{X_{min}}^{\otimes m}} : X_{min} \rightarrow \mathbb{P}^N$ is a morphism,
 $\omega_{X_{min}}^{\otimes m} \cong \phi^* \mathcal{O}_{\mathbb{P}^N}(1)$.
- It then follows that $R(\omega_X)$ is finitely generated.
- Conclusion: After performing a finite sequence of geometrically meaningful operations (contracting -1 curves) we obtain a birational model on which ω_X is positive (nef and semiample).

Birational geometry of surfaces V

- It should be stressed that it is impossible to classify all surfaces.
- If $\kappa(X) < 2$, we have a fairly good understanding of X .
- If $\kappa(X) = 2$, then we know that if $K_{X_{min}}^2 \leq M$ then there are only finitely many families of such surfaces (modulo birational isomorphism).
- It is however hopeless to classify all of these.

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3-folds

- The geometry of complex 3-folds ($d = 3$) was understood in the 1980's by work of S. Mori, Y. Kawamata, J. Kollár, M. Reid, V. Shokurov and others.
- They showed that if X is not covered by $\mathbb{P}^1$'s, then there exists a finite set of well understood birational maps (known as flips and divisorial contractions) $X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_N$ such that ω_{X_N} is semiample (i.e. $\phi_{\omega_{X_N}}^{\otimes m} \rightarrow \mathbb{P}^N$ is an embedding for some $m > 0$).
- In particular, $R(\omega_X)$ is finitely generated.

Higher dimensions

Recently there have been some very exciting developments in higher dimensions ($d \geq 4$). In particular we have:

Theorem (Birkar-Cascini-Hacon-M^cKernan)

Let X be a variety of general type (i.e. $\kappa(X) = \dim(X)$), Then there exists a finite sequence of flips and divisorial contractions

$$X \dashrightarrow X_1 \dashrightarrow \dots \dashrightarrow X_N$$

such that ω_{X_N} is semiample.

Higher dimensions II

We can also show that:

Theorem (Birkar-Cascini-Hacon-M^cKernan, Siu)

The canonical ring

$$R(\omega_X) = \bigoplus_{k \geq 0} H^0(X, \omega_X^{\otimes k})$$

is finitely generated.

It follows that there is a natural map $\phi : X \dashrightarrow \text{Proj} R(\omega_X) \subset \mathbb{P}_{\mathbb{C}}^m$.
 In most cases, ϕ carries a lot of information about X and it can be used to study the geometry of X in a coordinate free manner.

Higher dimensions III

One can also generalize some of the features of the classification of surfaces. For example:

Theorem (Tsuji, Hacon-M^cKernan, Takayama)

For any integers $d, M > 0$, there are only finitely many families of varieties of general type, dimension d and volume $\text{Vol}(\omega_X) = \omega_{X_{\min}}^d \leq M$ modulo birational isomorphisms.